

28 Supersymmetric Versions of the Standard Model

28.6 Gauge Mediation of Supersymmetry Breaking

In this section we will consider the possibility that the breakdown of supersymmetry is transmitted to the known particles through interactions of the ordinary $SU(3) \cdot SU(2) \cdot U(1)$ gauge bosons and their superpartners [30]. It is assumed here that supersymmetry is dynamically broken in a sector of superfields, *not* including the superfields of the observed quarks and leptons, and that some of the chiral superfields in the symmetry-breaking sector, known as the *messenger superfields*, have non-vanishing $SU(3) \cdot SU(2) \cdot U(1)$ quantum numbers. In order for the messenger particles to be able to get large (say, of order 1 TeV) masses without breaking $SU(3) \cdot SU(2) \cdot U(1)$, it is necessary for them to furnish a real (or pseudoreal) representation of $SU(3) \cdot SU(2) \cdot U(1)$, which automatically also means that they do not introduce any new anomalies. Although most treatments of gauge-mediated supersymmetry breaking in the literature also make specific assumptions about the interactions of the messenger superfields with the other superfields responsible for supersymmetry breaking, the most important predictions of this class of theories in fact do not depend on these assumptions. We shall therefore put off assuming anything about the interactions of the messenger superfields with the other superfields of the symmetry-breaking sector. We will, however, make another assumption about the $SU(3) \cdot SU(2) \cdot U(1)$ properties of the

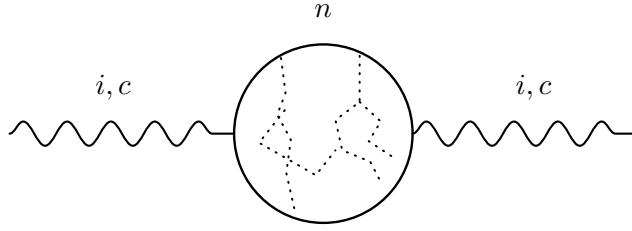


Figure 28.5. A diagram of the sort that introduces a breakdown of supersymmetry into the propagator of the gauge superfields. Here wavy lines are any component fields of the gauge superfields; solid lines are component fields of the messenger superfields; and dotted lines are component fields of the $SU(3) \cdot SU(2) \cdot U(1)$ -neutral superfields of the supersymmetry-breaking sector.

messenger superfields, which has a strong phenomenological motivation. In order for the messenger particles not to interfere with the unification of couplings discussed in Section 28.2, we assume that they have the same ratios for the total traces of all squared $SU(3) \cdot SU(2) \cdot U(1)$ gauge generators as the ordinary quarks and leptons. This condition will automatically be satisfied if the messenger superfields (together perhaps with some $SU(3) \cdot SU(2) \cdot U(1)$ -neutral chiral superfields) furnish a complete representation of some simple group G that contains $SU(3) \cdot SU(2) \cdot U(1)$, of which the quarks and leptons (again, together perhaps with some $SU(3) \cdot SU(2) \cdot U(1)$ -neutral chiral superfields) also form a complete representation. (For instance, these left-chiral superfields might form N $SU(2)$ singlet $SU(3)$ triplets with charge $e/3$ and N $SU(2)$ doublet $SU(3)$ singlets with charges 0 and $-e$, which together form N representations **5** of $SU(5)$, together with an equal number of left-chiral superfields in the complex-conjugate representations of $SU(3) \cdot SU(2) \cdot U(1)$,

which form N representations $\bar{\mathbf{5}}$.) However, for our present purposes we will neither need to assume that G is an actual symmetry group of the theory, nor adopt any particular choice of G or of the representations furnished by the messenger particles.

This interaction of the messenger superfields both with the other chiral and/or gauge superfields of the supersymmetry-breaking sector and with the $SU(3) \cdot SU(2) \cdot U(1)$ gauge superfields can be expected to produce a breakdown of supersymmetry in the propagators of the component fields of the $SU(3) \cdot SU(2) \cdot U(1)$ gauge superfields. To lowest order in the $SU(3) \cdot SU(2) \cdot U(1)$ couplings, the leading contribution to the propagators comes from the diagram shown in Figure 28.5, in which a pair of gauge, gaugino, or auxiliary D -field lines is attached to a loop of messenger fields, which also may have any number of interactions with $SU(3) \cdot SU(2) \cdot U(1)$ -neutral fields of the supersymmetry-breaking sector. The supersymmetry-breaking corrections Δ_{ic} to the propagators of the gauge superfields (with $i = 1, 2, 3$ for $SU(3)$, $SU(2)$, and $U(1)$, and $c = V, \lambda, D$ labelling the different components of each gauge superfield) therefore have the forms

$$\begin{aligned}\Delta_{3c}(q) &= (g_s^2/16\pi^2) \sum_n T_{3n} \Pi_{cn}(q), \\ \Delta_{2c}(q) &= (g^2/16\pi^2) \sum_n T_{2n} \Pi_{cn}(q), \\ \Delta_{1c}(q) &= (g'^2/16\pi^2) \sum_n T_{1n} \Pi_{cn}(q),\end{aligned}\tag{28.6.1}$$

where n labels the different messenger superfields; $\Pi_{cn}(q)$ are more-or-less complicated functions of the four-momentum q ; T_{3n} and T_{2n} are the traces of the squares of any generator of $SU(3)$ and $SU(2)$, respectively, in the representations furnished by the n th messenger superfield (normalized so that in the defining representations $T_3 = T_2 = 1/2$); and T_{1n} is the sum of the squares of the electroweak hypercharges of the n th messenger superfield. One immediate consequence is that the gauginos acquire masses of the same form:¹

$$\begin{aligned}m_{\text{gluino}} &= (g_s^2/16\pi^2) \sum_n T_{3n} M_{gn}, \\ m_{\text{wino}} &= (g^2/16\pi^2) \sum_n T_{2n} M_{gn}, \\ m_{\text{bino}} &= (g'^2/16\pi^2) \sum_n T_{1n} M_{gn},\end{aligned}\tag{28.6.2}$$

where the M_{gn} are masses that characterize the different messenger superfields. As already mentioned, in order to preserve the unification of couplings at very high energy we assume that the sums of the T_n have the same ratios as for the observed quarks and leptons:

$$\sum_n T_{3n} = \sum_n T_{2n} = \sum_n 3T_{1n}/5 = T.\tag{28.6.3}$$

¹Recall that the bino is the superpartner of the $U(1)$ gauge field B_μ that appears in the Lagrangian of the standard model. We are not yet taking $SU(2) \cdot U(1)$ breaking into account, so the gaugino, squark, and slepton masses calculated here should be understood as parameters appearing in the $SU(3) \cdot SU(2) \cdot U(1)$ -invariant effective Lagrangian of the standard mode.

The breakdown of supersymmetry in these propagators is then communicated to the squarks and sleptons of the supersymmetric standard model through the diagrams shown in Figure 28.6, in which an

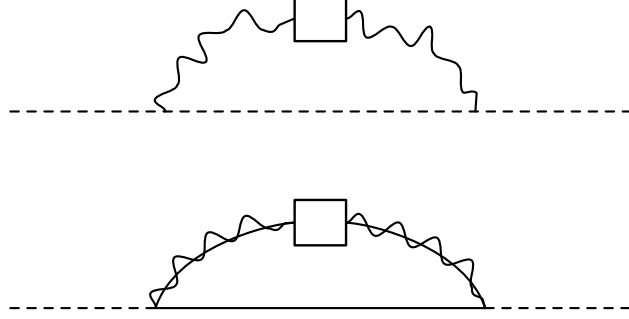


Figure 28.6. Diagrams that communicate supersymmetry breaking to the squarks and sleptons. Here dashed lines are squarks or sleptons; wavy lines are $SU(3) \cdot SU(2) \cdot U(1)$ gauge bosons or auxiliary D fields; solid lines are quarks or leptons; combined solid and wavy lines are $SU(3) \cdot SU(2) \cdot U(1)$ gauginos; and the squares represent insertions of the supersymmetry-breaking propagator correction shown in Figure 28.5.

$SU(3) \cdot SU(2) \cdot U(1)$ gauge boson or gaugino or auxiliary D field is emitted and re-absorbed by the squark or slepton. We are calculating the effective low-energy theory in which $SU(3) \cdot SU(2) \cdot U(1)$ -breaking is not yet taken into account, so there is no mixing among the $SU(3)$, $SU(2)$, and $U(1)$ propagators, and each propagator acts like a unit matrix on the gauge indices. Thus the squared mass given to any squark or slepton will be proportional to a sum over the squares of all the $SU(3) \cdot SU(2) \cdot U(1)$ generators (including coupling constants) in the representation furnished by that squark or slepton. The sums of the squared $SU(2)$ and $SU(3)$ generators in the defining representations are

$$\sum_{a=1}^3 (g\sigma_a/2)^2 = \frac{3g^2}{4} \mathbf{1}, \quad \sum_{\alpha=1}^8 (g_s\lambda_\alpha/2)^2 = \frac{4g_s^2}{3} \mathbf{1},$$

where σ_a are the Pauli isospin matrices (5.4.18) and λ_α are the Gell-Mann matrices (19.7.2). For $U(1)$ the generator is just the weak hypercharge (21.3.7), including a factor g' . The

squared squark and slepton masses therefore have the forms

$$\begin{aligned}
M_Q^2 &= 2 \sum_n M_{sn}^2 \left[\frac{4}{3} \left(\frac{g_s^2}{16\pi^2} \right)^2 T_{3n} + \frac{3}{4} \left(\frac{g^2}{16\pi^2} \right)^2 T_{2n} + \left(\frac{1}{6} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 T_{1n} \right], \\
M_{\bar{U}}^2 &= 2 \sum_n M_{sn}^2 \left[\frac{4}{3} \left(\frac{g_s^2}{16\pi^2} \right)^2 T_{3n} + \left(\frac{2}{3} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 T_{1n} \right], \\
M_{\bar{D}}^2 &= 2 \sum_n M_{sn}^2 \left[\frac{4}{3} \left(\frac{g_s^2}{16\pi^2} \right)^2 T_{3n} + \left(-\frac{1}{3} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 T_{1n} \right], \\
M_L^2 &= 2 \sum_n M_{sn}^2 \left[\frac{3}{4} \left(\frac{g^2}{16\pi^2} \right)^2 T_{2n} + \left(\frac{1}{2} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 T_{1n} \right], \\
M_{\bar{E}}^2 &= 2 \sum_n M_{sn}^2 \left(\frac{g'^2}{16\pi^2} \right)^2 T_{1n},
\end{aligned} \tag{28.6.4}$$

where Q , $\bar{U}$, $\bar{D}$, L , and $\bar{E}$ are the scalar superpartners of the left-handed quark doublets, the left-handed antiquarks of charge $-2e/3$ and $+e/3$, the left-handed lepton doublets, and the left-handed charged antileptons, and M_{sn} are some new masses that characterize the n th messenger superfields. (The factor 2 is extracted from M_{sn}^2 for future convenience.) The squark and slepton masses that are produced in this way are automatically the same in all three generations, thus avoiding the problem with flavor-changing processes discussed in Section 28.4.

We expect all M_{gn} and M_{sn} to be roughly of the same order of magnitude, so that the gluino and squarks will have comparable masses, while the wino, bino, and sleptons will be much lighter, with masses suppressed by squares of electroweak coupling constants.

We can go considerably further than this with some reasonable dynamical assumptions. Suppose that the effects of supersymmetry breaking on the messenger superfields may be modeled by including these superfields along with a set of $SU(3) \cdot SU(2) \cdot U(1)$ -neutral chiral superfields S_n (not necessarily all distinct) in a superpotential

$$f(\Phi, \bar{\Phi}, S) = \sum_n \lambda_n S_n \Phi_n \bar{\Phi}_n, \tag{28.6.5}$$

where $\bar{\Phi}_n$ and Φ_n are left-chiral messenger superfields in complex-conjugate representations of $SU(3) \cdot SU(2) \cdot U(1)$ and λ_n are a set of coupling coefficients. (Here and below we suppress the $SU(3) \cdot SU(2)$ indices that are summed over in calculating scalar products like $\Phi_n \bar{\Phi}_n$.) The superfields S_n are supposed to have non-vanishing vacuum expectation values $\mathcal{S}_n$ and $\mathcal{F}_n$ for their scalar and auxiliary components, respectively. It is the non-zero values of $\mathcal{F}_n$ that in these models introduce supersymmetry breaking in the masses of the Φ_n and $\bar{\Phi}_n$ particles. Section 26.4 shows that, with gauge couplings neglected, the squared masses of the spinor components of Φ_n (and $\bar{\Phi}_n$) are the eigenvalues of the matrix $\mathcal{M}_n^\dagger \mathcal{M}_n$, with $\mathcal{M}_n$ defined by Eq. (26.4.11), which gives

$$\mathcal{M}_n = \begin{pmatrix} 0 & \lambda_n \mathcal{S}_n \\ \lambda_n \mathcal{S}_n & 0 \end{pmatrix},$$

so that the messenger fermions have masses $|\lambda_n \mathcal{S}_n|$. To find the mass terms for the scalar components ϕ_n and $\bar{\phi}_n$ of the Φ_n and $\bar{\Phi}_n$ superfields, we note that integrating out the auxiliary fields of the Φ_n and $\bar{\Phi}_n$ yields a potential

$$\sum_n \left| \frac{\partial f(\phi, \bar{\phi}, \mathcal{S})}{\partial \phi_n} \right|^2 + \sum_n \left| \frac{\partial f(\phi, \bar{\phi}, \mathcal{S})}{\partial \bar{\phi}_n} \right|^2 = \sum_n |\lambda_n \mathcal{S}_n|^2 [|\phi_n|^2 + |\bar{\phi}_n|^2],$$

to which we now must add the contribution of the auxiliary component of the S_n , given by the second term of Eq. (26.4.4) as:

$$2 \operatorname{Re} \sum_n \left[\lambda_n \mathcal{F}_n \frac{\partial f(\phi, \bar{\phi}, \mathcal{S})}{\partial \mathcal{S}_n} \right] = 2 \operatorname{Re} \sum_n [\mathcal{F}_n \lambda_n \phi_n \bar{\phi}_n].$$

The complex scalar fields of definite mass are then $(\phi_n \pm e^{-i\alpha_n} \bar{\phi}_n)/\sqrt{2}$, where α_n is the phase of $\lambda_n \mathcal{F}_n$, with squared masses $|\lambda_n \mathcal{S}_n|^2 \pm |\lambda_n \mathcal{F}_n|$. (Note that this pattern, of a pair of complex scalars with squared masses equidistant above and below a Majorana fermion squared mass, is just what we would expect from the sum rule (27.5.11).) Since these squared masses must be positive, it follows that

$$|\mathcal{F}_n| \leq |\lambda_n| |\mathcal{S}_n|^2. \quad (28.6.6)$$

The gaugino masses in models based on Eq. (28.6.5) are given by diagrams of the form shown in Figure 28.5, but now with just a single loop, not including what are shown as dotted lines in Figure 28.5. A detailed calculation gives the coefficients M_{gn} in Eq. (28.6.2) as [31]

$$M_{gn} = \frac{|\mathcal{F}_n|}{|\mathcal{S}_n|} g \left(\frac{|\mathcal{F}_n|}{|\lambda_n| |\mathcal{S}_n|^2} \right), \quad (28.6.7)$$

where

$$\begin{aligned} g(x) &\equiv \frac{1}{2x^2} [(1+x) \ln(1+x) + (1-x) \ln(1-x)] \\ &= 1 + \frac{x^2}{6} + \frac{x^4}{15} + \frac{x^6}{28} + \dots \end{aligned} \quad (28.6.8)$$

The masses of the squarks and sleptons are given by the diagrams of Figure 28.6, which now involve just two loops. Another detailed calculation gives the mass parameters M_{sn}^2 in Eq. (28.6.4) as [31]

$$M_{sn}^2 = \frac{|\mathcal{F}_n|^2}{|\mathcal{S}_n^2|} f \left(\frac{|\mathcal{F}_n|}{|\lambda_n| |\mathcal{S}_n|^2} \right), \quad (28.6.9)$$

where

$$\begin{aligned} f(x) &= \frac{1+x}{x^2} \left[\ln(1+x) - 2 \operatorname{Li}_2 \left(\frac{x}{1+x} \right) + \frac{1}{2} \operatorname{Li}_2 \left(\frac{2x}{1+x} \right) \right] + x \longrightarrow -x \\ &= 1 + \frac{1}{36} x^2 - \frac{11}{450} x^4 - \frac{319}{11760} x^6 + \dots, \end{aligned} \quad (28.6.10)$$

with Li_2 the dilogarithm

$$\operatorname{Li}_2(x) \equiv - \int_0^x \frac{\ln(1-t)}{t} dt. \quad (28.6.11)$$

In particular, if (as is usually assumed) the various S_n are all the same, and if $|\mathcal{F}| \ll |\lambda_n||\mathcal{S}|^2$ for all n , then f and g in Eqs. (28.6.9) and (28.6.7) may be set equal to 1, so that

$$M_{gn} = M_{sn} = |\mathcal{F}|/|\mathcal{S}| \equiv M. \quad (28.6.12)$$

Using Eq. (28.6.3), we can express the gaugino masses (28.6.2) as

$$\begin{aligned} m_{\text{wino}} &= (g^2/16\pi^2)TM, \\ m_{\text{bino}} &= (5/3)(g'^2/16\pi^2)TM, \\ m_{\text{gluino}} &= (g_s^2/16\pi^2)TM, \end{aligned} \quad (28.6.13)$$

while the squared squark and slepton masses (28.6.4) become

$$\begin{aligned} M_Q^2 &= 2TM^2 \left[\frac{4}{3} \left(\frac{g_s^2}{16\pi^2} \right)^2 + \frac{3}{4} \left(\frac{g^2}{16\pi^2} \right)^2 + \frac{5}{3} \left(\frac{1}{6} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 \right], \\ M_U^2 &= 2TM^2 \left[\frac{4}{3} \left(\frac{g_s^2}{16\pi^2} \right)^2 + \frac{5}{3} \left(\frac{2}{3} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 \right], \\ M_D^2 &= 2TM^2 \left[\frac{4}{3} \left(\frac{g_s^2}{16\pi^2} \right)^2 + \frac{5}{3} \left(-\frac{1}{3} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 \right], \\ M_L^2 &= 2TM^2 \left[\frac{3}{4} \left(\frac{g^2}{16\pi^2} \right)^2 + \frac{5}{3} \left(\frac{1}{2} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 \right], \\ M_E^2 &= 2TM^2 \frac{5}{3} \left(\frac{g'^2}{16\pi^2} \right)^2. \end{aligned} \quad (28.6.14)$$

There is no special reason to expect that $|\mathcal{F}| \ll |\lambda_n\mathcal{S}|^2$, but this assumption is not actually very restrictive, because Eq. (28.6.6) already requires that $|\mathcal{F}| \leq |\lambda_n||\mathcal{S}|^2$, and it turns out that the functions $f(x)$ and $g(x)$ do not differ much from unity for $x < 1$ unless x is very close to 1.

The extreme simplicity of the results (28.6.13) and (28.6.14) has been explained by Giudice and Rattazzi [32] by using Seiberg's arguments of holomorphy [33], described in Section 27.6. Suppose we introduce a superpotential for the messenger superfields like (28.6.5), but with a single external singlet superfield S :

$$f(S, \Phi) = S \sum_n \lambda_n \Phi_n \bar{\Phi}_n. \quad (28.6.15)$$

Among other effects the kinematic term for the gauge superfield V_i (with $i = 3, 2, 1$ for $SU(3)$, $SU(2)$, and $U(1)$) in the Wilsonian effective Lagrangian at a renormalization scale μ will now take the form

$$\mathcal{L}_{\text{gauge}, \mu} = \text{Re} \left[\sum_i N_i(S, \mu) \sum_{\alpha\beta} (W_{iL\alpha} \epsilon_{\alpha\beta} W_{iL\beta}) \right]_{\mathcal{F}}, \quad (28.6.16)$$

with some functions $N_i(S, \mu)$ replacing the factors $1/2g_i^2(\mu)$ in Eq. (27.3.22). (The θ -term is dropped here because it has no effect in perturbation theory. A sum is implied over indices

on $W_{iL\alpha}$ that label different members of the adjoint representations of $SU(3)$ and $SU(2)$, which are not explicitly shown.) The gauge coupling constant is now given by setting the superfield S equal to the expectation value $\mathcal{S}$ of its scalar component

$$\frac{1}{2g_i^2(\mu)} = N_i(\mathcal{S}, \mu). \quad (28.6.17)$$

Also, recalling that $W_{iL\alpha} = \lambda_{iL\alpha} + O(\theta)$ and using Eq. (27.2.11), the terms in the Lagrangian density (28.6.16) of second order in the gaugino fields are, up to derivative terms,

$$-2 \sum_i \text{Re} [N_i(\mathcal{S}, \mu) (\bar{\lambda}_{iR} \not{\partial} \lambda_{iR}) + [N_i(S, \mu)]_{\mathcal{F}} (\lambda_{iL}^T \epsilon \lambda_{iL})].$$

This gives the gaugino masses

$$m_{gi}(\mu) = \left| \frac{[N_i(S, \mu)]_{\mathcal{F}}}{2N_i(\mathcal{S}, \mu)} \right| = g_i^2(\mu) |[N_i(S, \mu)]_{\mathcal{F}}|. \quad (28.6.18)$$

Now let us consider the behavior of $N_i(\mathcal{S}, \mu)$ as a function of a real positive $\mathcal{S}$, with the phases of the messenger superfields adjusted so that all λ_n are real and positive. Suppose we fix the values of the gauge couplings $g_i(\mu)$ at some scale $\mu = K$ above all the messenger particle masses. Taking account of the change in the constants b_i in the renormalization group equations $\mu dg_i(\mu)/d\mu = b_i g_i^3$ as μ passes through the various messenger masses, the solution of this equation when μ is *below* all the messenger particle masses takes the form

$$\frac{1}{g_i^2(\mu)} = \frac{1}{g_i^2(K)} - 2b_i^{(0)} \ln \left(\frac{M_1}{K} \right) - 2b_i^{(1)} \ln \left(\frac{M_2}{M_1} \right) - \dots - 2b_i^{(N)} \ln \left(\frac{\mu}{M_N} \right),$$

where we label the messenger particles so that their masses $M_n = \lambda_n \mathcal{S}$ satisfy

$$M_1 > M_2 > \dots > M_N,$$

and $b_i^{(n)}$ is calculated taking account only of particles with masses less than M_n . Since all M_n are proportional to $\mathcal{S}$, we see that $N_i(\mathcal{S}, \mu)$ has the $\mathcal{S}$ dependence

$$N_i(\mathcal{S}, \mu) = -b_i^{\text{messenger}} \ln \mathcal{S} + \mathcal{S}\text{-independent terms}, \quad (28.6.19)$$

where $b_i^{\text{messenger}} = b_i^{(0)} - b_i^{(N)}$ is the contribution to b_i of all the messenger superfields. According to Eq. (27.9.45) (with $C_{i1} = 0$ and $C_{i2}^f = C_{i2}^s = \sum_n T_{in}$), this is

$$b_i^{\text{messenger}} = \frac{1}{16\pi^2} \sum_n T_{in}. \quad (28.6.20)$$

Since $N_i(S, \mu)$ is required by supersymmetry to be a holomorphic function of S , we see that

$$N_i(S, \mu) = -\frac{1}{16\pi^2} \sum_n T_{in} \ln S + S\text{-independent terms}. \quad (28.6.21)$$

Expanding around $S = \mathcal{S}$, to first order in $\mathcal{F}$ we have $[\ln S]_{\mathcal{F}} = \mathcal{F}/\mathcal{S}$ and so Eq. (28.6.18) gives the gaugino masses

$$m_{gi}(\mu) = \frac{g_i^2(\mu)}{16\pi^2} \sum_n T_{in} \left| \frac{\mathcal{F}}{\mathcal{S}} \right|. \quad (28.6.22)$$

Using Eq. (28.6.3), we see that this is the same as our previous result (28.6.13). Eq. (28.6.14) for the squark and slepton masses was obtained by Giudice and Rattazzi in a similar way, by studying the kinematic terms for the quark and lepton superfields instead of the gauge superfields.

Incidentally, Eqs. (28.6.13) and (28.6.14) could be obtained (in general with different values of M in Eq. (28.6.13) and Eq. (28.6.14)) without a specific dynamical assumption like Eq. (28.6.5), if we supposed that the supersymmetry-breaking sector respected invariance under some grand unified group G that had both the known quarks and leptons and the messenger fields as complete representations. In this case the coefficients M_{gn} and M_{sn} in Eqs. (28.6.2) and (28.6.4) would have values $M_g(d)$ and $M_s(d)$, respectively, that depend only on the irreducible representation d of G to which the n th messenger fields belong. The sums of T_{in} over n belonging to any irreducible representation d of G have the same ratios as the sums in Eq. (28.6.3), so $\sum_{n \in d} T_{in} = k_i T(d)$, where $k_3 = k_2 = 1$, $k_1 = 5/3$, and therefore

$$\sum_n T_{in} M_{gn} = \sum_d M_g(d) \sum_{n \in d} T_{in} = k_i M_g,$$

where $M_g = \sum_d M_g(d) T(d)$. Likewise

$$\sum_n T_{in} M_{sn}^2 = \sum_d M_s^2(d) \sum_{n \in d} T_{in} = k_i M_s^2,$$

where $M_s^2 = \sum_d M_s^2(d) T(d)$. Eqs. (28.6.2) and (28.6.4) would then yield Eqs. (28.6.13) and (28.6.14), except with M_g in place of TM and M_s^2 in place of $2TM^2$. But the assumption that M_{gn} and M_{sn} respect invariance under G is implausible if the messenger mass scale is far below the grand unified scale, since whatever we assume about a grand unified gauge group, the $SU(3) \cdot SU(2) \cdot U(1)$ gauge interactions would make coupling constants like λ_n run differently for Φ_n with different $SU(3) \cdot SU(2) \cdot U(1)$ quantum numbers in the same representation of the grand unified group.

These results are subject to various radiative corrections, of which the most important is that we must use values of g_s , g , and g' renormalized at a scale comparable to the mass being calculated. Indeed, the gaugino mass ratios given in Eq. (28.6.13) could also be derived under quite different assumptions: that all gaugino masses are equal at the grand unification scale where the coupling constants are related by $g_s^2 = g^2 = 5g'^2/3$ and become different at lower energies as described by the renormalization group equations.

For an illustrative example of numerical results, suppose that the messenger superfields form an $SU(2)$ singlet $SU(3)$ triplet with charge $e/3$ and an $SU(2)$ doublet $SU(3)$ singlet with charges 0 and $-e$, together with left-chiral superfields in the complex-conjugate representations of $SU(3) \cdot SU(2) \cdot U(1)$. Then, as already mentioned, Eq. (28.6.3) is satisfied with $T = 2 \cdot 1/2 = 1$, so by using the correct values of the gauge couplings, the masses of the squarks, gluino, L sleptons, wino, E sleptons, and bino are calculated to be in the ratios [34] $11.6 :: 7.0 :: 2.5 :: 2 :: 1.1 :: 1.0$. In larger representations of G we could have $T \gg 1$, in which case the gluino would be the heaviest of these particles, and the sleptons the lightest ones.

There are also radiative corrections aside from the running of the gauge couplings. According to one calculation [34], in the model with an $SU(2)$ singlet $SU(3)$ triplet with charge $e/3$ and an $SU(2)$ doublet $SU(3)$ singlet with charges 0 and $-e$, together with left-chiral superfields in the complex-conjugate representations of $SU(3) \cdot SU(2) \cdot U(1)$, radiative corrections give the ratios of the masses of the squarks, gluino, L sleptons, wino, E sleptons, and bino as $9.3 :: 6.4 :: 2.6 :: 1.9 :: 1.35 :: 1.0$.

As we saw in the previous section, the wino and bino may mix with the charged and neutral higgsinos, so the wino and bino masses that have been calculated here must be regarded as inputs to a calculation of physical masses of mixtures known as *charginos* and *neutralinos*, rather than physical masses themselves.

Now we shall consider the masses of the Higgs scalars in these models. If we were to consider only the two-loop diagrams by which these scalars get masses through gauge interactions with the supersymmetry-breaking sector, then since (apart from signs) they have the same $SU(3) \cdot SU(2) \cdot U(1)$ quantum numbers as the left-handed lepton doublets, their masses would be given by a formula like the fourth of Eqs. (28.6.4)

$$\begin{aligned} [m_1^2]_{2 \text{ loop}} &= [m_2^2]_{2 \text{ loop}} = M_L^2 \\ &= \sum_n M_{sn}^2 \left[\frac{3}{4} \left(\frac{g^2}{16\pi^2} \right)^2 T_{2n} + \left(\frac{1}{2} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 T_{1n} \right]^2. \end{aligned} \quad (28.6.23)$$

If this were the whole story, then it would be impossible to meet the condition for $SU(2) \cdot U(1)$ breaking found in the previous section that (unless $\tan \beta$ is very close to unity) one of $m_1^2 + |\mu|^2$ and $m_2^2 + |\mu|^2$ must be negative. Fortunately, the large masses of the top quark and squark produce a negative contribution to m_2^2 that leads naturally to a spontaneous breakdown of electroweak symmetry. The couplings of the Higgs doublets to the third-generation quark superfields are described by the superpotential

$$f_{3\text{rd gen}} = \lambda_b (H_1^T e Q) \bar{B} + \lambda_t (H_2^T e Q) \bar{T}, \quad (28.6.24)$$

where Q is the $SU(2)$ quark doublet left-chiral superfield (T, B) , while $\bar{T}$ and $\bar{B}$ are the left-chiral superfields of the left-handed top and bottom antiquarks, and λ_t and λ_b are the Yukawa couplings, related to the t and b quark masses by $m_t = \lambda_t v_2$ and $m_b = \lambda_b v_1$. The last term of Eq. (26.4.7) then gives the terms in the potential involving an interaction between the squark fields and Higgs fields as

$$\begin{aligned} V_{\text{sq } H} &= |\lambda_b \mathcal{H}_1^- \bar{\mathcal{B}} + \lambda_t \mathcal{H}_2^0 \bar{\mathcal{T}}|^2 + |\lambda_b \mathcal{H}_1^0 \bar{\mathcal{B}} + \lambda_t \mathcal{H}_2^+ \bar{\mathcal{T}}|^2 \\ &\quad + |\lambda_b|^2 |\mathcal{H}_1^0 \mathcal{B} - \mathcal{H}_1^- \mathcal{T}|^2 + |\lambda_t|^2 |\mathcal{H}_2^+ \mathcal{B} - \mathcal{H}_2^0 \mathcal{T}|^2. \end{aligned} \quad (28.6.25)$$

where script letters denote the scalar field components of superfields. The squark loop contributions to the potential of $\mathcal{H}_1$ and $\mathcal{H}_2$ are then

$$V_H^{\text{squark loop}} = 3 \langle \mathcal{S} \mathcal{S}^* \rangle \left[2 |\lambda_b|^2 (\mathcal{H}_1^\dagger \mathcal{H}_1) + 2 |\lambda_t|^2 (\mathcal{H}_2^\dagger \mathcal{H}_2) \right], \quad (28.6.26)$$

where $\langle \mathcal{S} \mathcal{S}^* \rangle$ is the vacuum expectation value of the product of any one of the squark fields and its complex conjugate at the same spacetime point. (In taking this to be the same

for all squark types, we are here using Eq. (28.6.4), which tells us that the squark masses M_Q do not vary much among $\mathcal{T}$, $\mathcal{B}$, $\bar{\mathcal{T}}$, and $\bar{\mathcal{B}}$ squarks. The factor 3 in Eq. (28.6.26) takes account of the three colors of each squark type.) The vacuum expectation value $\langle \mathcal{S}\mathcal{S}^* \rangle$ is given in lowest order by

$$\langle \mathcal{S}\mathcal{S}^* \rangle \equiv \langle \mathcal{S}(x)\mathcal{S}^*(x) \rangle_{\text{VAC}} = \frac{-i}{(2\pi)^4} \int \frac{d^4p}{p^2 + M_Q^2 - i\epsilon}. \quad (28.6.27)$$

This of course is divergent, but the contribution to supersymmetry-breaking coefficients would be cancelled by quark loops if the squarks had the zero bare mass of the quarks, so the effect of quark loops is to subtract from (28.6.27) the same expression with M_Q replaced with zero, which with a Wick rotation becomes

$$\begin{aligned} \langle \mathcal{S}\mathcal{S}^* \rangle &\longrightarrow \frac{M_Q^2 i}{(2\pi)^4} \int \frac{d^4p}{(p^2 + M_Q^2 - i\epsilon)(p^2 - i\epsilon)} \\ &= -\frac{M_Q^2}{(2\pi)^4} \int_0^{M^2} \frac{\pi^2 dp^2}{p^2 + M_Q^2} \simeq -\frac{M_Q^2}{16\pi^2} \ln \left(\frac{M^2}{M_Q^2} \right). \end{aligned}$$

We have inserted an ultraviolet cut-off at the messenger mass M , because at momenta above M the squark mass must be replaced with a momentum-dependent mass, which goes to the supersymmetric value zero at very high momenta. Making this substitution in Eq. (28.6.26) gives a net contribution to the potential due to squark and quark loops:

$$(V_m)^{\text{3 loop}} = \frac{3M_Q^2}{16\pi^2} \ln \left(\frac{M^2}{M_Q^2} \right) \left[2|\lambda_b|^2 (\mathcal{H}_1^\dagger \mathcal{H}_1) + 2|\lambda_t|^2 (\mathcal{H}_2^\dagger \mathcal{H}_2) \right]. \quad (28.6.28)$$

(This is a three-loop contribution, because the squared squark masses are given by two-loop diagrams. There are also terms in the potential that are quadratic in both the Higgs and squark superfields, arising from products of the Higgs and squark terms in the squares of the $SU(2) \cdot U(1)$ gauge field D -components. These do not make a three-loop contribution to the Higgs masses because the sum of each $SU(2) \cdot U(1)$ quantum number of the squarks vanishes, so that their contributions to the $SU(2) \cdot U(1)$ gauge field D -components have zero vacuum expectation value.) Comparing Eq. (28.6.28) with Eq. (28.5.9) and adding the two-loop contribution (28.6.23) to the masses, we see then that

$$m_1^2 \simeq M_L^2 - \frac{3M_Q^2 |\lambda_b|^2}{8\pi^2} \ln \left(\frac{M^2}{M_Q^2} \right), \quad (28.6.29)$$

$$m_2^2 \simeq M_L^2 - \frac{3M_Q^2 |\lambda_t|^2}{8\pi^2} \ln \left(\frac{M^2}{M_Q^2} \right). \quad (28.6.30)$$

Using Eq. (28.6.14) and $|\lambda_t| = m_t/v_2 = m_t(2\sqrt{2}G_F)^{1/2}/\sin\beta$, we may write (28.6.30) as

$$\begin{aligned} m_2^2 \simeq 2TM^2 &\left[\frac{3}{4} \left(\frac{g^2}{16\pi^2} \right)^2 + \frac{5}{12} \left(\frac{g'^2}{16\pi^2} \right)^2 \right. \\ &\left. - \frac{\sqrt{2}G_F m_t^2}{\pi^2 \sin^2 \beta} \left(\frac{g_s^2}{16\pi^2} \right)^2 \ln \left(\frac{3}{8T(g_s^2/16\pi^2)} \right) \right]. \end{aligned} \quad (28.6.31)$$

For $T = 1$, $g_s^2/4\pi = 0.118$, $g^2/4\pi = 0.0340$, $g'^2/4\pi = 0.0101$, and $m_t = 180$ GeV, this is

$$m_2^2 \simeq M_L^2 \left[1 - \frac{3.06}{\sin^2 \beta} \right], \quad (28.6.32)$$

which is negative for all values of β , thus providing a natural mechanism for the spontaneous breaking of the electroweak gauge symmetry. Also, $M_L^2 = (0.91 \cdot 10^{-4}) M^2 / 8\pi^2$. Unless $\tan \beta$ is huge, we have $|\lambda_b| \ll |\lambda_t|$, so Eq. (28.6.27) gives

$$m_1^2 \simeq M_L^2. \quad (28.6.33)$$

The aspect of electroweak phenomenology for which the predictions of gauge-mediated supersymmetry models are most uncertain and unsatisfactory has to do with the parameter μ in the supersymmetry-preserving term $\mu[(H_1^T e H_2)]_{\mathcal{F}}$ and the related supersymmetry-breaking term $B\mu$ in the Lagrangian density. These are related because the interactions of the Higgs superfields with the gauge, lepton, and quark superfields are invariant under a symmetry

$$\begin{aligned} H_1 &\longrightarrow e^{i\varphi} H_1, & H_2 &\longrightarrow e^{i\varphi} H_2, \\ Q &\longrightarrow e^{-i\varphi} Q, & V_i &\longrightarrow V_i, \\ \bar{D} &\longrightarrow \bar{D}, & \bar{U} &\longrightarrow \bar{U}, \end{aligned} \quad (28.6.34)$$

which in the absence of a superpotential term $\mu(H_1^T e H_2)$ would forbid radiative corrections from producing a term $B\mu \text{Re}(\mathcal{H}_1^T e \mathcal{H}_2)$ in the scalar field potential.

It is not possible for $B\mu$ to vanish because then Eq. (28.5.22) and the fact that (as we have seen) $m_A \neq 0$ would imply that $\sin 2\beta = 0$, or in other words either $v_1 = 0$ or $v_2 = 0$, which would imply that either all the charge $-e/3$ quarks and charged leptons are massless or all the charge $+2e/3$ quarks are massless. (If $B\mu = 0$ and $\mu = 0$ then this problem persists to all orders, because the appearance of non-zero vacuum expectation values for both v_1 and v_2 would imply that symmetry under any combination of the transformation (28.6.34) and electroweak $U(1)$ gauge transformations is spontaneously broken, so the C -odd neutral scalar would be a Goldstone boson with $m_A = 0$.) It is natural to try to account for a non-zero value of $B\mu$ as a radiative correction in a theory in which symmetry under the transformation (28.6.34) is explicitly broken by a supersymmetric term $\mu[(H_1^T e H_2)]_{\mathcal{F}}$ in the Lagrangian density. This gives a very small value of $B\mu$ at the messenger scale [34], although renormalization group effects greatly increase $B\mu$ at lower energies. According to Eq. (28.5.22), a relatively small $B\mu$ would fit well with the idea that the large mass of the top quark arises from a large value of $\tan \beta$. In any case, as already mentioned, the experimental lower bound on the chargino mass tells us that $|\mu|$ is at least about 60 GeV.

The trouble is that the appearance of a non-zero value of μ resurrects the hierarchy problem that supersymmetry was supposed to solve: instead of asking why the Higgs mass terms in the Lagrangian density are so much smaller than the Planck mass or the mass at which gauge couplings are unified, we now have to ask why μ is this small?

The hierarchy problem would be put to rest if the Higgs superfields interact with the supersymmetry-breaking sector in such a way that a term $\mu[(H_1^T e H_2)]_{\mathcal{F}}$ is forbidden by some symmetry, but appears when that symmetry is spontaneously broken. A massless

Goldstone boson may be avoided if the symmetry is discrete rather than continuous. The simplest possibility is simply to extend the symmetry transformation (28.6.34) to include a transformation

$$S \longrightarrow e^{-2i\varphi} S,$$

which would allow a term in the superpotential of the form

$$\lambda' S (\mathcal{H}_1^T e \mathcal{H}_2).$$

We can avoid a continuous symmetry by also including a term S^3 in the superpotential, so that the Lagrangian is invariant only under transformations with φ a multiple of $2\pi/3$, which is still enough to forbid a non-zero bare value of μ . In this case the appearance of non-zero vacuum values $\mathcal{S}$ and $\mathcal{F}$ for the scalar and auxiliary components of S yields

$$B\mu = |\lambda' \mathcal{F}|, \quad \mu = |\lambda' \mathcal{S}|.$$

This would result in B having a very large $|\lambda'|$ -independent value M , given by (28.6.12), which is greater by a factor of order $(g_s^2/16\pi)^{-1} \simeq 100$ than the squark or gluino masses. But then, since Eqs. (28.6.32) and (28.6.33) give $m_1^2 + m_2^2 < 0$, the stability condition (28.5.11) would require that $|\mu| \geq M/2$, and hence Eq. (28.5.22) would require that m_A is also much greater than the squark and slepton masses. This is ruled out by the relation (28.5.23) and the estimates (28.6.33) and (28.6.32) of m_1^2 and m_2^2 unless $\tan \beta$ is very close to unity.

We will see in Section 31.6 that theories of gravity-mediated supersymmetry breaking naturally yield acceptable values of $B\mu$ and μ . Such theories are characterized by a very high energy scale of supersymmetry breaking, in various versions of order either 10^{11} GeV or 10^{13} GeV. There have been several suggestions [35] on ways to obtain acceptable values of $B\mu$ and μ in theories in which supersymmetry is broken at relatively low energy, such as theories of gauge-mediated supersymmetry breaking, but none of them is particularly compelling. Also, since we do not know where μ comes from, we do not have any reason to suppose that it is real, so theories of gauge-mediated supersymmetry breaking are at risk of yielding too much CP violation, just as in the more general framework of Section 28.4.

Like the scalar squared masses m_1^2 and m_2^2 , the parameters A_{ij} and C_{ij} in Eq. (28.4.1) are given by two-loop diagrams. However, they have the dimensions of mass rather than mass squared, and they are much less than the scalar and gaugino masses, so they make a relatively unimportant contribution to supersymmetry breaking.

As in any model with supersymmetry broken at energies much less than 10^{10} GeV, the lightest R -odd particle in all models based on gauge-mediated supersymmetry breaking is the gravitino. As we will see in Section 31.3, the gravitino mass is of the order of $\sqrt{G}$ times the squared energy F characteristic of supersymmetry breaking, defined so that the vacuum energy is $F^2/2$. Where supersymmetry is broken by the $\mathcal{F}$ -terms $\mathcal{F}_{n0}$ of the $SU(3) \cdot SU(2) \cdot U(1)$ -neutral chiral superfields S_n , we have $F^2 = \sum_n |\mathcal{F}_{n0}|^2$. If there are no large dimensionless quantities in the Lagrangian for the S_n , then squark masses in this model are of the order of $g_s^2 \sqrt{F}/16\pi^2 \simeq 10^{-2} \sqrt{F}$, so in order for this to be less than the naturalness bound of 10^4 GeV, we must have $\sqrt{F} < 10^6$ GeV, which gives a gravitino mass

less than 1 keV. Gravitational couplings are so weak at accessible energies that it is only the helicity $\pm 1/2$ states of the gravitino that can actually be produced, and these behave just like goldstino states. As shown by Eq. (27.5.12), in the models discussed here the goldstino field appears with coefficient $i\sqrt{2}\mathcal{F}_n$ in the fermionic component ψ_n of S_n . Goldstinos are emitted in the decay of the R -odd sparticles into the corresponding R -even particles of the standard model through radiative corrections, with goldstinos emerging from vertices connecting internal ψ_n and S_n lines. According to Eq. (29.2.10), the goldstino emission amplitudes are inversely proportional to F , which makes these decays comparatively slow, although perhaps fast enough to be observed.

Because the decay of R -odd particles into goldstinos is slow, it is phenomenologically important in these models to identify the next-to-lightest R -odd particle, into which all the heavier R -odd particles will decay before it in turn decays into a goldstino. As we have seen, the next-to-lightest R -odd particle is usually a slepton, wino, or bino. (It is possible in models with a messenger superfield having the same $SU(3) \cdot SU(2) \cdot U(1)$ quantum numbers as the Higgs doublets that the mixing between these superfields would lower the mass of the doublet messengers so much that the lightest R -odd particle would be the gluino [35a].) Detailed calculations including the effects of $SU(2) \cdot U(1)$ breaking indicate that here is a large region of parameter space in which the next-to-lightest R -odd particle is one of the two tau sleptons [35b].